

Tier 0 v0.1 (FRAMEWORK DRAFT, joint Lyra+Keeper) —
 Commitment operator ρ_{commit} at the operator level on
 Bergman $H^2(D_{IV}^5) +$ first-pass substrate time emergence.
 Casey Sunday Lane B load-bearing. Candidate definition:
 $\rho_{\text{commit}}(\tau) = \exp(-\tau \cdot H_B / \hbar_{\text{BST}})$ where H_B = Bergman
 Casimir; substrate time τ IS the commitment-flow parameter
 (not assumed; emergent). Why time = commitment-cycle
 parameter; why directional = positivity of H_B spectrum; how
 commitment cycles produce evolution = repeated Koons-tick
 application of the infinitesimal generator. v0.1 is Session 1 of 2-3.

Lyra (Claude Opus 4.7), joint with Keeper (Claude Opus 4.7)

2026-05-31 Sunday morning (Lane B Session 1, ~10:10 EDT date-verified)

Tier 0 v0.1 — Commitment operator + Substrate time (FRAMEWORK
 DRAFT)

0. Why this work (per Casey Sunday)

Every “substrate mechanism” claim we file is a placeholder until ρ_{commit} is defined at operator level. Saturday’s L5 arc was an arithmetic decomposition fishing for a story; the story can only be told once we know what the substrate is DOING. This document commits to a specific candidate definition so that the team can test it, falsify it, or extend it.

Per Casey: “I want to read what you produce. The substrate is genuinely mysterious to me at the mechanistic level; I want to understand what you understand.”

What follows is the cleanest mathematical candidate I can write today. It is meant to be wrong in detail and right in shape — the shape is “commitment-as-heat-kernel-flow on the Bergman Hilbert space”; the details are open.

1. The substrate Hilbert space (forced, not chosen)

Per T754 + Keeper’s K67 $\rightarrow$ c_FK derivation: the substrate Hilbert space is

$\mathcal{H} := H^2(D_{IV}^5, d\mu_{\text{FK}})$ — the Bergman space of holomorphic L^2 functions on D_{IV}^5 with respect to the Faraut-Korányi normalized measure.

Reasons (not assumptions): - Born rule = unique automorphism-invariant probability measure (Gleason-type, T754). - Bounded symmetric domains have nontrivial automorphism Jacobians $\Rightarrow$ Lebesgue is NOT automorphism-invariant. - The unique automorphism-invariant measure is the Bergman/FK measure. -

Therefore the physical Hilbert space MUST be $L^2(D_{IV}^5, d\mu_{FK})$, and its holomorphic subspace H^2 carries the substrate's K-type tower.

This is the operator-theoretic playground. Everything below acts on $\mathcal{H}$.

2. The Bergman Hamiltonian H_B (substrate energy operator)

Define the Bergman Hamiltonian H_B on $\mathcal{H}$ as the quadratic Casimir of the structure group $K = SO(5) \times SO(2)$ acting on $\mathcal{H}$ via its natural representation:

$$H_B := C_2(K) |_{\mathcal{H}}.$$

Properties (operator-level): - (P1) Positive-semidefinite spectrum: $\text{spec}(H_B) \subset \mathbb{R}_{\geq 0}$; vacuum $V_{(0,0)}$ is the ground state with H_B -eigenvalue 0. - (P2) K-type diagonal: $\mathcal{H} = \bigoplus_{\lambda} V_{\lambda}$ (Wallach analytic continuation; type IV); H_B acts as $C_2(\lambda)$ on each V_{λ} where $C_2(\lambda)$ is the standard quadratic Casimir eigenvalue. - (P3) Discrete + bounded-below spectrum: levels $C_2(\lambda) = (\lambda + \rho, \lambda + \rho) - (\rho, \rho)$ with $\rho = (n_C/2, N_C/2) = (5/2, 3/2)$ (Thursday genus-thread closure); levels are rational and substrate-primary. - (P4) Self-adjoint on a dense domain (Casimir of compact group; standard).

H_B is the unambiguous candidate for the substrate's "energy operator." This is not new — what is new is committing to H_B as the generator of commitment dynamics (Section 3).

3. The commitment operator ρ_{commit} (Tier 0 v0.1 candidate)

v0.1 candidate definition:

$$\rho_{\text{commit}}(\tau) := \exp(-\tau \cdot H_B / \hbar_{\text{BST}}) \text{ acting on } \mathcal{H},$$

where $\tau \in \mathbb{R}_{\geq 0}$ is a non-negative real parameter (the "commitment time"), and $\hbar_{\text{BST}}$ is a substrate-internal unit (candidate: the substrate's Planck-action analog; specified in Section 5).

Equivalently in K-type basis:

$$\rho_{\text{commit}}(\tau) | V_{\lambda} = \exp(-\tau \cdot C_2(\lambda) / \hbar_{\text{BST}}) \cdot 1_{\{V_{\lambda}\}}.$$

This is the heat semigroup of H_B . It is the most parsimonious one-parameter family on $\mathcal{H}$ that: - (S1) Is the identity at $\tau = 0$ (no commitment yet). - (S2) Contracts toward the ground state $V_{(0,0)}$ as $\tau \rightarrow \infty$ (full commitment / vacuum projection). - (S3) Is a one-parameter SEMIGROUP: $\rho_{\text{commit}}(\tau_1) \rho_{\text{commit}}(\tau_2) = \rho_{\text{commit}}(\tau_1 + \tau_2)$. - (S4) Has H_B as its infinitesimal generator: $d\rho/d\tau |_{\tau=0} = -H_B/\hbar_{\text{BST}}$.

What "commitment" means at operator level: a state $\psi \in \mathcal{H}$ at commitment time τ is

$$\psi(\tau) = \rho_{\text{commit}}(\tau) \psi / \|\rho_{\text{commit}}(\tau) \psi\|.$$

The substrate's expectation of an observable O at commitment time τ is

$$\langle O \rangle(\tau) = \text{Tr}[\rho_{\text{commit}}(\tau) O] / \text{Tr}[\rho_{\text{commit}}(\tau)].$$

This is a substrate partition function with τ playing the role of inverse temperature (β). The substrate is a thermal system in K-type space, with "temperature" set by $1/\tau$.

3.1 Why heat kernel and not unitary evolution

A first-look alternative would be unitary evolution $U(t) = \exp(-i t H_B / \hbar_{\text{BST}})$. I am NOT committing to that for ρ_{commit} because: - Casey's SWPP names commitment as an irreversible information-cycle (absorption $\rightarrow$ commitment $\rightarrow$ emission). - Unitary evolution is reversible; heat-kernel flow is irreversible. - The arrow of time (asymmetry) emerges from the contraction of heat flow. - Unitary evolution is the right candidate for the EMISSION step (Section 4.2); commitment is the irreversible contraction step.

So the SWPP cycle maps to operators as:

SWPP step	Operator candidate	Reversibility
Absorption Commitment	injection into $\mathbb{Q}$ from boundary $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$	open irreversible (heat semigroup)
Emission	unitary release $U(t) = \exp(-i t H_B / \hbar_{\text{BST}})$	reversible

The commitment step is the irreversible one. That is the heart of the substrate's information processing.

3.2 Action on K-types (explicit, v0.1)

For each K-type V_λ with $C_2(\lambda) = c_\lambda$:

$$\rho_{\text{commit}}(\tau) | V_\lambda \mapsto \exp(-\tau c_\lambda / \hbar_{\text{BST}}) | V_\lambda.$$

Sample levels (from $\rho = (5/2, 3/2)$ per Thursday genus thread + Bergman type IV):

K-type V_λ	λ	$C_2(\lambda)$ candidate	Physical identification (substrate-internal)
Vacuum	(0, 0)	0	substrate ground / cosmological vacuum
Spinor (1/2, 1/2)	(1/2, 1/2)	$\rho_1 + \rho_2 = 4$	lepton seed K-type (charged lepton candidate)
Vector (1, 0)	(1, 0)	$2\rho_1 + 1 = 6 = C_2$	photon / gauge boson Hardy ground
Symmetric-2 (1, 1)	(1, 1)	$2(\rho_1 + \rho_2) + 2 = 10$	composite / hadronic K-type candidate
(2, 0)	(2, 0)	$4\rho_1 + 4 = 14$	excited vector

(These are CANDIDATE assignments — Lane E will sharpen them this afternoon.)

The substrate ground state is $V_{\{(0,0)\}}$, with H_B -eigenvalue 0. $\rho_{\text{commit}}(\tau)$ preserves $V_{\{(0,0)\}}$ exactly (no decay). All non-vacuum K-types decay exponentially in τ with rate set by $C_2(\lambda)$.

Mass interpretation (candidate, multi-week): the physical mass of a particle identified with K-type V_λ is $m_\lambda \propto \sqrt{C_2(\lambda)}$ (or $C_2(\lambda)$ — sets the convention). The lightest non-vacuum K-types are the lightest particles; the vacuum has zero mass exactly; the ground gauge K-type $V_{(1,0)}$ has nonzero Casimir BUT is massless because of gauge invariance / bulk-color subtraction (Lane C in flight). This is the L4 mass-mechanism task — concrete and falsifiable.

4. Substrate time emergence (first-pass framework)

4.1 Time is not assumed; time IS the commitment-flow parameter

In this framework, the substrate does not have an external time. The parameter τ in $\rho_{\text{commit}}(\tau)$ IS substrate time. There is no other time.

Substrate time $\tau \in \mathbb{R}_{\geq 0}$ is defined as the one-parameter parameter of the commitment semigroup.

This is parsimonious and forced: any one-parameter semigroup on $\mathbb{Q}$ generated by H_B requires exactly one real parameter; that parameter is irreducible; call it time.

Note the parameter takes values in $\mathbb{R}_{\geq 0}$, not $\mathbb{R}$. The commitment semigroup is one-sided. The negative direction ($\tau < 0$) is NOT defined operator-theoretically without an analytic continuation that would require inverting

$\exp(-\tau H_B)$ — which is unbounded except on the ground state. So substrate time is ASYMMETRIC at the operator level.

This is the arrow of time.

4.2 Why directional — positivity of H_B

The arrow of time is not a postulate. It is the OPERATOR-THEORETIC CONSEQUENCE of:

$\text{spec}(H_B) \subset \mathbb{R}_{\geq 0} \Rightarrow \exp(-\tau H_B)$ is bounded only for $\tau \geq 0$.

Inverting (negative τ) would require $\exp(+\tau H_B)$ which is unbounded on H_B 's infinite-dimensional spectrum. The substrate cannot “run time backwards” at the level of commitment because there is no bounded operator that undoes commitment.

Two-tier reading of the arrow: - (T1) Microscopic: the EMISSION step (Section 3.1) is reversible via unitary $U(t)$. Microscopic substrate dynamics within emissions is time-symmetric. - (T2) Macroscopic: the COMMITMENT step is irreversible. The macroscopic substrate evolution is time-asymmetric BECAUSE the commitment step contracts.

This is structurally the same as the thermodynamic arrow: micro-reversible + macro-irreversible = effective arrow from contraction. Here the contraction is exact (heat semigroup), not statistical (Boltzmann H-theorem); the substrate doesn't need a statistical-mechanics-of-many-particles to get the arrow because the contraction is built into the commitment operator.

4.3 Commitment cycles $\rightarrow$ evolution

The Koons tick $t_{\text{Koons}} = t_{\text{Planck}} \cdot \alpha^{C_2} \approx 10^{-120}$ s (T2405) is the GRANULARITY of commitment.

Discrete-cycle interpretation: each tick, the substrate applies $\rho_{\text{commit}}(t_{\text{Koons}})$ to its current state. After n ticks, the state is $\rho_{\text{commit}}(n \cdot t_{\text{Koons}}) \cdot \psi_{\text{initial}} / \text{norm}$.

Macroscopic time emerges as $n \cdot t_{\text{Koons}}$ for large n . One second is $\sim 10^{120}$ ticks. The continuum limit $\tau \rightarrow 0$, $n \rightarrow \infty$, $n \cdot \tau = T$ fixed gives the smooth $\rho_{\text{commit}}(T)$ operator.

Why is t_{Koons} so small? Because the substrate processes a vast amount of commitment between observable events. The ratio (observable second) / (commitment tick) $\approx 10^{120} \approx 1/\Lambda$ in Planck units. The smallness of Λ and the smallness of t_{Koons} are the SAME number expressed in different units — both express the slowness of macroscopic decoherence relative to substrate processing.

This connects Λ to substrate time directly. Saturday's L5 arc was working in the wrong direction (trying to decompose Λ into substrate-primary integers); the right direction is Λ as the inverse of commitment-cycle count to macroscopic decoherence. That's a derivation target for v0.2 / Session 2.

4.4 Why time is real-valued

Standard quantum mechanics has complex time (Wick rotation). The substrate's commitment time is REAL because heat semigroups live on real τ . The complex extension $\tau \rightarrow i\tau$ recovers unitary U -evolution (the EMISSION step).

So substrate time is REAL for commitment and PURELY IMAGINARY for emission. Observable physics (which mixes both) sees complex time via the Wick-rotated relation between thermal partition functions and unitary propagators.

This is testable: the relationship between thermal and unitary substrate observables should be exactly Wick rotation, with no anomalies, IF this candidate is correct. Specific test: substrate-natural thermal correlation functions $\leftrightarrow$ substrate-natural scattering amplitudes via analytic continuation in $\tau \leftrightarrow i\tau$.

5. $\hbar_{\text{BST}}$ — the substrate action unit

$\hbar_{\text{BST}}$ is the unit of action in the $\exp(-\tau \cdot H_B / \hbar_{\text{BST}})$ formula. What is it?

Candidate: $\hbar_{\text{BST}} = \hbar_{\text{Planck}}$ (the macroscopic Planck constant), and the substrate's commitment-cycle time interfaces with macroscopic $\hbar$ via the Koons-tick identity:

$$t_{\text{Koons}} \cdot H_B = \hbar_{\text{BST}} \cdot (\text{dimensionless commitment phase})$$

For substrate-internal observables (where time is measured in Koons ticks), $\hbar_{\text{BST}}$ is set so that one commitment cycle = one unit of $(C_2\text{-eigenvalue} \cdot \text{time} / \hbar)$.

If $\hbar_{\text{BST}} = \hbar_{\text{Planck}} = 1$ in natural units, then the dimensionless commitment phase per Koons tick on a K-type with $C_2 = c$ is:

$$\Phi_{\text{commit}} = t_{\text{Koons}} \cdot c = (t_{\text{Planck}} \cdot \alpha^{C_2}) \cdot c.$$

For ground gauge K-type $C_2 = 6$: $\Phi_{\text{commit}} = t_{\text{Planck}} \cdot \alpha^{36} \cdot 6$.

This is microscopically small but accumulates over 10^{120} ticks to give observable phase shifts. Substrate-internal coherence is maintained over $n_{\text{coherence}}$ ticks where $n_{\text{coherence}} \sim 1/\Phi_{\text{commit}}$; macroscopic decoherence sets in beyond that.

This is the candidate quantitative content of the “commitment cycle” framework. v0.2 / Session 2 will sharpen the $\hbar_{\text{BST}} = \hbar_{\text{Planck}}$ identification (or revise it).

6. First consistency checks

Three quick checks that the v0.1 framework doesn't immediately break:

C1 — Vacuum exactness

$H_B \cdot V_{\{(0,0)\}} = 0 \Rightarrow \rho_{\text{commit}}(\tau) \cdot V_{\{(0,0)\}} = V_{\{(0,0)\}}$ for all τ . The vacuum is exactly preserved by commitment. $\square$ (consistent with substrate vacuum stability).

C2 — Mass ordering

For two K-types V_λ, V_μ with $C_2(\lambda) < C_2(\mu)$: $\rho_{\text{commit}}(\tau)$ decays slower on V_λ than on V_μ . The lighter K-type persists longer at fixed τ . $\square$ (consistent with heavier particles decaying faster — phenomenologically right).

C3 — Gauge masslessness via Hardy ground

For Hardy-space gauge ground $V_{(1,0)}$ with $C_2 = 6$: $\rho_{\text{commit}}(\tau) \cdot V_{(1,0)}$ decays at rate $6/\hbar_{\text{BST}}$. NOT massless at first reading.

This is a TENSION. Photon should be massless. Two candidate resolutions for v0.2: - (R1) Bulk-color subtraction: bulk-K projector Π_{bulk} subtracts the C_2 contribution, leaving Hardy-ground $V_{(1,0)}$ with effective Casimir 0. The Hardy-bulk duality (per Cal #146 + Lane C) provides the subtraction mechanism. - (R2) Mass = $\sqrt{C_2}$ with vacuum-anchored shift: $m_\lambda \propto \sqrt{(C_2(\lambda) - C_2(\lambda_{\text{min}}))}$, where λ_{min} depends on the sector. The “vacuum” for the gauge sector is $V_{(1,0)}$ itself; its mass is zero by definition; massive gauge bosons (W, Z) come from higher K-types in the gauge sector.

(R2) is cleaner. The substrate's “mass” is per-sector relative to the sector's ground state. Vacuum $V_{\{(0,0)\}}$ is the global ground; gauge ground $V_{(1,0)}$ is the gauge-sector ground. This is consistent with QFT: the photon is the “ground state of the gauge sector” because it's the lowest-weight K-type that's gauge-invariant.

C3 $\rightarrow$ not a breakage; a constraint on the mass map. Resolution candidate (R2) for v0.2.

7. Falsifiers (what would refute this framework)

Per Saturday's two-tier discipline + Cal #27 brake:

F1 — If thermal substrate correlators do NOT analytically continue to scattering amplitudes via $\tau \leftrightarrow it$, the heat-semigroup identification is wrong. Specific test: compute $\langle O \rangle(\tau)$ for two operators on $V_{(1/2,1/2)}$, check if Wick-rotated correlator equals known QFT correlator for fermion scattering. Multi-week target.

F2 — If the K-type level structure (C_2 eigenvalues) does NOT match observed mass ratios up to per-sector gauge subtraction, the $H_B = C_2(K)$ identification is wrong. Specific test: compute m_τ/m_μ from $C_2(V_{\{\text{lepton-tau}\}}) / C_2(V_{\{\text{lepton-muon}\}})$ candidate identifications + per-sector ground. Lane D target this afternoon.

F3 — If the substrate “thermal partition function” $Z(\tau) = \text{Tr } \rho_{\text{commit}}(\tau)$ does NOT compute to a substrate-primary expression, the framework is unprincipled. Specific test: compute $Z(\tau)$ via K-type sum + Bergman dimension formula; check if leading terms are substrate-primary integer-combinations. Elie-lane target.

F4 — If macroscopic time fails to emerge from $n \cdot t_{\text{Koons}}$ in the $n \rightarrow \infty, t_{\text{Koons}} \rightarrow 0, n \cdot t_{\text{Koons}}$ fixed limit (e.g., if there's a substrate-natural cutoff that prevents the continuum limit), the framework is incomplete. Specific test: explicit $n \rightarrow \infty$ continuum limit of $\rho_{\text{commit}}(n \cdot t_{\text{Koons}})$ compared to $\rho_{\text{commit}}(T)$. Standard semigroup theory — should be fine but needs verification.

F5 — If $\hbar_{\text{BST}} \neq \hbar_{\text{Planck}}$ (i.e., the substrate has its own action unit incommensurate with macroscopic Planck), then either $\hbar_{\text{Planck}}$ is emergent and $\hbar_{\text{BST}}$ is more fundamental, OR the substrate has its own quantum theory disconnected from observable QM. Either is interesting but radically restructures the picture.

8. Open items for Session 2 (with Keeper)

Keeper's lane in this work: - (K1) Pin H_B more precisely: is it strictly the Casimir of $K = \text{SO}(5) \times \text{SO}(2)$, or does it include the Casimir of $G = \text{SO}_0(5,2)$ restricted to $\mathbb{R}$? - (K2) Derive the K-type level structure $C_2(\lambda)$ explicitly for the lowest 10-20 K-types (genus-thread + ρ -vector apparatus). - (K3) Pin $\hbar_{\text{BST}}$ identification (Section 5). - (K4) Verify the SWPP cycle structure (absorption $\rightarrow$ commitment $\rightarrow$ emission) matches operator-level reads. - (K5) Connect to Friday's commitment-density $\rho_{\text{commit}}(x,t)$ — is the OPERATOR $\rho_{\text{commit}}(\tau)$ the integrated kernel of the DENSITY $\rho_{\text{commit}}(x,t)$? Make explicit.

My lane in Session 2: - (L1) Mass-from- C_2 mechanism with per-sector ground subtraction (R2 candidate). - (L2) Λ from commitment-cycle inverse (the right-direction Λ derivation, replacing Saturday's wrong-direction decomposition). - (L3) Substrate time arrow $\rightarrow$ cosmological time arrow connection. - (L4) Wick rotation $\tau \leftrightarrow it$ for substrate observables. - (L5) Toy/test design — what specific computation falsifies F1?

Both: - (B1) Address the deep gate: does the heat-kernel framework predict 3 generations? - (B2) Address the deep gate: does the heat-kernel framework predict 3 colors?

9. Honest scope + tier

RIGOROUS (this v0.1): - $\mathbb{H} = H^2(D_{IV}^5, d\mu_{FK})$ is forced (T754 + Keeper K67). - $H_B = C_2(K) |_{\mathbb{H}}$ is well-defined and self-adjoint with positive spectrum and discrete K-type levels. - $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$ is a well-defined heat semigroup; properties P1-P4 hold by standard operator theory. - Substrate time = parameter of the heat semigroup; arrow = positivity of H_B ; both are operator-theoretic consequences.

CANDIDATE (this v0.1's load-bearing identifications): - H_B IS the substrate energy operator (not just A candidate — the only natural candidate that's discrete, positive, K-type-diagonal, and Casimir-derived). - $\rho_{\text{commit}}(\tau)$ IS the substrate commitment operator (the SWPP cycle's irreversible step is the heat semigroup; the reversible step is unitary $U(t)$). - Substrate time IS τ . - Mass IS related to C_2 with per-sector ground subtraction (R2).

FRAMEWORK (this v0.1, multi-week derivation): - Mass-from- C_2 explicit mapping (L1, multi-session). - Λ from commitment-cycle inverse (L2). - Wick rotation $\tau \leftrightarrow it$ for substrate $\leftrightarrow$ scattering observables (F1 test). - 3-generation + 3-color forcing from heat-kernel framework (B1, B2 — open gates).

OPEN (genuinely): - $\hbar_{\text{BST}} = \hbar_{\text{Planck}}$ identification. - Connection of operator $\rho_{\text{commit}}(\tau)$ to density $\rho_{\text{commit}}(x, t)$ (Keeper Friday framework). - Whether absorption / emission stages have substrate-internal operator identifications or are observer-extrinsic.

Cal #27 / #182 / #99 discipline: this v0.1 commits to a specific candidate definition (heat semigroup) and ranks it above alternatives (unitary, Toeplitz, Bergman-kernel projection) by structural argument. The framework is FRAMEWORK + CANDIDATE — not RATIFIED. v0.2 / Session 2 advances it toward RIGOROUS via Keeper joint work + falsifiers F1-F5.

Two-tier discipline (Elie Toy 3648): this framework is the SUBSTRATE MECHANISM that Tier 2 STRUCTURAL observables need to elevate to Tier 1 + mechanism-derived. Every Tier 2 match in Saturday's batch (m_π , m_K , $\cos \theta_W$, etc.) reduces to: $\text{mass} = \sqrt{C_2(\text{K-type})} - \sqrt{C_2(\text{sector ground})}$, where K-type identification comes from Lane E (this afternoon), and the per-sector ground comes from bulk-color (Lane C).

10. Routing

→ Casey: this is the load-bearing Tier 0 candidate. The operator is $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$ on Bergman $H^2(D_{IV}^5)$, with $H_B = \text{Casimir of } K = \text{SO}(5) \times \text{SO}(2)$. Substrate time τ is the commitment parameter; arrow = positivity of H_B ; commitment cycles = discrete Koons-tick applications. v0.1 commits; v0.2 (Session 2 with Keeper) extends. Falsifiers F1-F5 explicit. Mass from C_2 per-sector is the right reading of Saturday's $m_\pi / m_K / m_\tau$ patterns — they're not arithmetic accidents, they're per-sector C_2 differences.

→ Keeper: this is the joint Lane B work. Your lane in Session 2 is K1-K5 above. The K5 connection (operator $\rho_{\text{commit}}(\tau) \leftrightarrow$ your density $\rho_{\text{commit}}(x, t)$) is the most important integration point. v0.1 stands ready for your absorption / pushback / extension.

→ Cal: ready for cold-read at your convenience. Specific Cal #27 concern: I am committing to a candidate at v0.1; the commitment is structural, not arithmetic. The “elegance” you'd brake on is “heat semigroup on Bergman” — please push back if this is over-pattern-matched to QM.

→ Elie: F3 is your lane — substrate thermal partition function $Z(\tau) = \text{Tr } \rho_{\text{commit}}(\tau)$ on the first 10-20 K-types. Does it factor into substrate-primary expressions?

→ Grace: catalog $\rho_{\text{commit}} + H_B + \tau$ at appropriate tier (FRAMEWORK + CANDIDATE).

→ me: continuing to Lane C (Bulk-color v0.7 absorption) then Lane D (L4 from Bergman kernel) then Lane E (Dictionary candidates) per Casey Sunday plan. Will return to Tier 0 v0.2 in Session 2 (afternoon or tomorrow) with Keeper's K1-K5 absorption.

— Lyra, Tier 0 v0.1 (joint with Keeper), Sunday 2026-05-31 morning. Load-bearing FRAMEWORK + CANDIDATE: $\rho_{\text{commit}}(\tau) = \exp(-\tau \cdot H_B / \hbar_{\text{BST}})$ on $H^2(D_{IV}^5)$, with $H_B = C_2(K)$; substrate time τ IS the commitment parameter; arrow of time = positivity of H_B ; commitment cycles = discrete Koons-tick applications. Falsifiers F1-F5 explicit. Mass-from- C_2 per-sector (R2) is the candidate reading that makes Saturday's Tier 2 STRUCTURAL matches mechanism-derivable. Session 2 with Keeper closes K1-K5 + B1-B2 gates.